SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

FACULTY OF ENGINEERING AND TECHNOLOGY

SCHOOL OF COMPUTING



COURSE PLAN

21CSC305P- MACHINE LEARNING

JULY – NOVEMBER 2026

Revision History:

Date	Version	Modification done	Modified by	Reviewed by	Authorized by
17.07.2026	1.0	Initial Release	Dr S SHANMUGAM	Dr. Murugan	
20.07.2026	2.0				

Table of Contents

1.0 General Details....3
2.0 Reference Books....3
3.0 Prerequisites....3
4.0 Instructional Objectives....3
5.0 Overall Assessment Plan....4
6.0 Tentative Test Schedule....4
7.0 Detailed Test Plan....5
8.0 List of practice experiments....5
9.0 Mini Project Details....6
10.0 Quiz/Puzzles/Scenario Based Assignments....6
11.0 Course Certification....7
12.0 Detailed Session Plan....7
13.0 Overall Execution Plan....9

1.0 General Details

Course Code: 21CSC305P

Course Title: MACHINE LEARNING

Semester: V

Course Time: JULY – NOVEMBER 2026

Slot: F

Day	Batch			
	Batch 1		Batch 2	
	Hour	Timing	Hour	Timing
Day order 1	3,4	09.45 - 11.30	8,9	2.20 - 4.00
Day order 2	-	-	-	-
Day order 3	-	-	-	-
Day order 4	-	-	-	-
Day order 5	4	10.40 -11.30	9	3.10 - 4.00

Location: Tech Park/University Building

2.0 Reference Books

1. Pattern Recognition and Machine Learning, Christopher M. Bishop,Springer, 2006
2. Machine Learning: a Probabilistic Perspective, Kevin Patrick Murphy. MIT Press, 2012

3.0 Prerequisites

Nil

4.0 Instructional Objectives

1. Explore the fundamental mathematical concepts of machine learning algorithms
2. Apply linear machine learning models to perform regression and classification
3. Utilize mixture models to group similar data items
4. Develop machine learning models for time-series data prediction
5. Design ensemble learning models using various machine learning algorithms

5.0 Overall Assessment Plan

S. No	Components		Marks	Portal Mark Entry Date
1	FP-I	CT1 (Unit 1 – Poster Presentation (5 Marks) – Chart based activity &Unit 2 – Mini Coding Task (5 marks)) **Practical Challenge**	10	24-9-2026
2	FP-II	CT2 (Unit-3 & 4 – **Written Test**)	10	28-10-2026
3	PBL-I	Project Review-1	5	12-10-2026
		Practice Experiments Demonstration and Viva (1 to 4)	10	
		Interactive Class Room Activities Research Paper Review & Kaggle Mini Challenge	5	
4	PBL-II	Project Review-2	5	13-11-2026
		Discover & Analyze ML Algorithms – Infographic Creation (Videos - Animations videos)	5	
		Practice Experiments Demonstration and Viva (5 to 9)	10	
5	PBL-III	Project Review-3	5	18-11-2026
		CT3- **Knowledge Check (**Quiz)	5	
		Certification course (NPTEL) **12 Weeks Course-**Introduction to Machine Learning – IIT Madras Certification course (NPTEL) **Exam Date: 18 OCT 2026 IST**	10	
6	R21 Report and Viva	Project Report (Max. 30 pages)	15	18-11-2026
		Viva Voce (5 marks)	5	
TOTAL			100	

6.0 Tentative Test Schedule-

S. No	Tentative date	Test	Marks	Portion	Duration
1	18-08-2026 (Poster) & 10-09-2026(Coding)	Cycle Test-1 **Practical Challenge**	50	Unit- 1 &2	100 mins
2	12-10-2026	Cycle Test-2 **Written Test**	50	Unit – 3 & 4	100 mins
3	03-11-2026	Cycle Test-3 **Knowledge Check** (Quiz)	25	Unit – 5	50 mins

7.0 Detailed Test Plan

Test	Tentative Date	Type	Marks	Mode
Cycle Test – I	18-08-2026 (Poster) & 10-09-2026(Coding)	**Practical Challenge**	Total: 50 Marks Chart Based Activity – 25 Marks (Unit 1) Mini Coding Challenge – Unit 2 (25 Marks)	Physical Exam
Cycle Test – II	12-10-2026	**Written Test**	Total: 50 Marks Exam Pattern: MCQs - 10 Concept Understanding Questions – 4 out of 6 (4*5) Scenario based / HOTs Questions - 2 (Either -or Choice) (2*10)	Physical Exam
Cycle Test – III	03-11-2026	**Knowledge Check** (Quiz)	Total: 25 Marks Exam Pattern: 25 Questions (25*1)	Physical Exam

8.0 List of Practice Experiments:

S. No	List of practice experiments	Dates
1	Devise a program to import, load, and view a dataset.	**Exp 1 to 4 02-09-2026**
2	Create a program to display the summary and statistics of the dataset.	
3	Implement linear regression to perform prediction.	
4	Implement Bayesian logistic regression and SVM for classification.	
5	Implement K-means clustering, mixtures of Gaussians, and hierarchical clustering algorithm to categorize data.	**Exp 5 to 9 12-10-2026**
6	Create a program to perform PCA.	
7	Implement HMM to predict the sequential data.	
8	Implement CART learning algorithms to perform categorization.	
9	Implement ensemble learning models to perform classification.	

9.0 Mini Project Details

S. No	Project Components	Dates	Evaluation method	Description
1	Project Review 1 (Marks -5)	11.08.26	Understanding the problem statement – 2.5 Marks Statistical analysis of dataset – 2.5Marks	Pertaining to unit-1 specific problem statements for a Batch of 4 students will be assigned as per the problem statements identified by the respective faculty members of the class.
2	Project Review 2 (Marks -5)	10.09.26	Mathematical Modelling of the algorithm-2Marks Development of python script for the mathematical model - 3Marks	Pertaining to Unit-2 & 3, as per the algorithms taught in the class, the problem statement chosen by the team must draft the mathematical model for the problem and associated python script. (avoid python libraries)
3	Project Review 3 (Marks -5)	04.11.26	Demo -3 Marks Comparative Analysis -2 Marks	Pertaining to Unit-4 & 5, the problem chosen must be executed and the result has to be compared with the algorithms taught in Unit- 4 & 5. Github link has to be submitted for plagiarism check.
4	Report, Viva (Marks -15)	11.11.26	Report-10 Marks Viva – 5 Marks	A detailed project report has to be prepared for the problem statement, objectives, mathematical model, written python script, result with comparative analysis.

				The student's knowledge on understanding the subject concepts will be evaluated through the presentation and his/her answering skills during the time of viva.

10.0 Interactive Learning Split-up Plan

Test	Tentative date of evaluation	Marks	Split-up
PBL-I Interactive Class Room Activities	11.08.26	5 Marks	**Research Paper Review: Each team reviews one recent ML paper.** Presentation (5 Marks) Problem Description (4 Marks) Dataset and Algorithm (4 Marks) Results and Discussion (4 Marks) Advantages & Limitations (4 Marks) Future work (4 Marks) Total: 25 Marks (Scale down to **6 marks**) (**Note: The team composition will be the same as that of the Mini Project teams**) **Kaggle Mini Challenge** Students are required to develop a machine learning model to predict the target variable using the provided training dataset. Train a model (5 Marks) Improve accuracy(5 Marks) Submit predictions(5 Marks) (**Note: The team composition will be the same as that of the Mini Project teams**)
PBL-II Discover & Analyze ML Algorithms – Infographic Animations videos Team of Four Students	10.09.26	5 Marks	Conceptual Clarity (5 marks) Visual Communication & Design (5 marks) Technical Content & Accuracy (5 marks) Total: 15 Marks (Scale down to **5 marks**) Video to be submitted **(Note: The team composition will be the same as that of the Mini Project teams**)

11.0 Course Certification- NPTEL (ONLINE- Introduction to Machine Learning - IITM - Duration (July – November 2026)

S. No	NPTEL Certification Split-up	Marks
1	>=50% and consistent Assignment submission (**10 marks**)	10
2	<50% and consistent Assignment submission (**4 marks**)	
3	Not Enrolled (**0 marks**)	

NPTEL Course:

1. **12 Weeks Course-**Introduction to Machine Learning – IIT Madras Certification course (NPTEL) **Exam Date: 18 OCT 2026 IST. https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs133**

12.0 Detailed Session Plan

HOURS	TOPICS	BOOK	TEACHING METHOD	TESTING METHOD
UNIT-1 INTRODUCTION				
1	Machine learning- What and Why	B2	BB/PPT	Illustration using example
2	Supervised Learning, Unsupervised learning	B2	BB/PPT	Illustration using example
3	Polynomial curve fitting	B1	BB/PPT	Illustration using example
4	Probability theory- discrete random variables, Fundamental rules, Bayes rule	B2	BB/PPT	Illustration using example
5	Independence and conditional independence	B2	BB/PPT	Illustration using example
6	Continuous random variables, Quantiles, mean and variance	B2	BB/PPT	Illustration using example
7	Probability densities, Expectation and covariance	B1	BB/PPT	Illustration using example
8	Practice 1: Devise a program to import, load, and view a dataset. Practice 2: Create a program to display the summary and statistics of the dataset.		Demo	
9	Statistical Analyzer Poster			
UNIT-2 LINEAR MODELS FOR REGRESSION				
10	Maximum likelihood estimation- Least squares	B2	BB/PPT	Illustration using example
11	Robust linear expression, ridge regression, Bayesian linear regression	B2	BB/PPT	Illustration using example
12	Linear models for classification: Discriminant function	B1	BB/PPT	Illustration using example
13	Probabilistic generative models, Probabilistic discriminative models	B1	BB/PPT	Illustration using example
14	Laplacian approximation, Bayesian logistic regression	B1	BB/PPT	Illustration using example
15	Kernel functions, using kernels in GLM, Kernel trick	B2	BB/PPT	Illustration using example
16	SVMs	B2	BB/PPT	Illustration using example
17	Practice 1:Implement linear regression to perform prediction. Practice 2:Implement Bayesian logistic regression and SVM for classification.		Demo	
18	Discover & Analyze ML Algorithms – Infographic Creation (Animations videos)			
UNIT-3 MIXTURE MODELS AND EM				
19	K-means clustering	B1	BB/PPT	Illustration using example
20	Mixtures of Gaussians	B1	BB/PPT	Illustration using example
21	Alternative view of EM	B1	BB/PPT	Illustration using example

22	Factor analysis	B2	BB/PPT	Illustration using example
23	PCA, Choosing the number of latent dimensions	B2	BB/PPT	Illustration using example
24	Clustering- measuring dissimilarity, Evaluating the output of clustering methods	B2	BB/PPT	Illustration using example
25	Hierarchial clustering	B2	BB/PPT	Illustration using example
26	Practice 1:Implement K-means clustering, mixtures of Gaussians, and hierarchical clustering algorithm to categorize data. Practice 2: Create a program to perform PCA.		Demo	
27	Discover & Analyze ML Algorithms – Infographic Creation (Animations videos)			
UNIT- 4 HIDDEN MARKOV MODELS				
28	Sequential data- Markov models	B1	BB/PPT	Illustration using example
29	HMM	B1	BB/PPT	Illustration using example
30	Maximum likelihood for the HMM	B1	BB/PPT	Illustration using example
31	The forward and backward algorithm	B1	BB/PPT	Illustration using example
32	The sum-product algorithm	B1	BB/PPT	Illustration using example
33	Scaling factors, Viterbi algorithm	B1	BB/PPT	Illustration using example
34	Linear dynamical systems	B1	BB/PPT	Illustration using example
35	Practice 1: Implement HMM to predict the sequential data.		Demo	
36	Discover & Analyze ML Algorithms – Infographic Creation (Animations videos)			
UNIT- 5 COMBINING MODELS				
37	Bayesian model averaging	B1	BB/PPT	Illustration using example
38	Boosting	B1	BB/PPT	Illustration using example
39	Adaptive basis function models	B2	BB/PPT	Illustration using example
40	Classification and regression trees (CART)	B2	BB/PPT	Illustration using example
41	Generalized additive models	B2	BB/PPT	Illustration using example
42	Ensemble learning	B2	BB/PPT	Illustration using example
43	Practice 1: Implement CART learning algorithms to perform categorization.		Demo	
44	Practice 2: Implement ensemble learning models to perform classification.		Demo	
45	Quiz			

13. Overall Execution Plan:

#	Activity	Target Dates	Responsibilities	Assigned to
1	PPT Content Preparation	31-07-26	Individual faculty can have their own course materials following the given reference books	All faculties.
2	Question Bank Preparation	15-08-26	1. Each faculty to prepare for the respective units assigned. 2. Questions must be framed on own and not to be taken as such from any other source. Other sources can be referred, but the question must be modified, say with different example program, and so on. 3. Solution is required for all questions. Concept Understanding Questions - 4 Scenario based / HOTs Questions - 2 4. Team Heads are responsible for distributing topics to team members and no topics are missed.	All faculties
4	Question Bank Scrutiny	16-08-26	1. Check for the quality of the questions as per the category in the question bank. 2. Ensure there are no repetitions. 3. Coordinate with CC.	SPOC Team
5	Cycle Test	24-10-26	1. Select the question from Question Bank 2. Share the QP to audit professor for review 3. Plan for cycle tests question paper printing, print and distribute. 4. Coordinate with CC.	SPOC Team
6	Course File Preparation	10-11-26	1. Responsible for the preparation of course file as per the checklist. 2. At the end of each CT exam, files should be updated and got verified from the Team Head. 3. Participate in result analysis activity. 4. Course Files are to be prepared for each department and the faculties listed are responsible for the preparation including CO-PO Mapping, attainment of Cos, etc. 5. Coordinate with CC.	SPOC, Course File Team
8	Mini Project	15-07-26	1. Number of students per team – 4. 2. Ensure no repetition of projects in the class. 3. Conduct periodic reviews to evaluate progress. 4. Verify research papers during the review.	All faculties
9	Feedback Collection and Minutes of Meeting	13-11-26	1. Scribe and prepare minutes of meeting for all meetings conducted. 2. Share the MoM to CC and Audit professors on the same day or the next of meeting.	SPOC Team